

Articles for Literature Review

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1 Methods for Fairness Assessment and Interpretability

1. Mehrabi et al. (2021). Fairness in Machine Learning: A Survey. [Link]

This paper identifies three main categories of fairness interventions: pre-processing approaches, which modify training data; in-processing approaches; and post-processing approaches, which adjust model outputs. The survey reviews widely used fairness metrics including demographic parity, equalized odds, and predictive parity—and discusses their trade-offs, such as balancing fairness with accuracy. This work also dives in the definition of the word "fair" providing plenty of various meanings.

2. Rudin et al. (2021). Interpretable Machine Learning: Fundamental Principles and 10 Grand Challenges. [Link]

This paper identifies 10 “grand challenge” technical problems in the field. The authors argue against relying on post-hoc explanations for black box models, emphasizing that such explanations can be misleading and reinforce false trust. Instead, they advocate designing models that are interpretable by construction. The survey clarifies that “interpretability” is often confused with “explainability,” and it seeks to separate these concepts.

3. Caton & Haas (2020). Bias Mitigation for Machine Learning Classifiers: A Comprehensive Survey. [Link]

This is a large literature review of 341 articles on methods to reduce bias and improve fairness in ML classifiers. They also classify all bias mitigation methods into pre-processing, in-processing and post-processing. The survey identifies common datasets used to evaluate fairness and popular metrics.

4. Mehrabi et al. (2023). A Critical Survey on Fairness Benefits of Explainable AI. [Link]

Authors collected 175 research papers from various fields that talk about how explainable AI (XAI) could make AI systems more fair, analyzed each paper to see what claims the authors made about the link between XAI and fairness, and discovered that many papers either don't clearly define which type of fairness they mean or don't back their statements by evidence. This study also uses the definitions of "formal fairness" and "perceived fairness".

5. Novielli et al. (2024). SoK: Taming the Triangle – Fairness, Interpretability and Privacy. [Link]

This is a Systematization of Knowledge (SoK) study that examines how fairness, inter-

pretability, and privacy interact, support, or conflict with each other. For instance, adding fairness constraints can require revealing sensitive data, impacting privacy. They propose “co-optimization” strategies, grouping methods into: Fairness-Privacy approaches, Fairness-Interpretability approaches, Interpretability-Privacy approaches, and triple-constraint approaches that try to balance all three of the previous strategies.

2 Reviews of Resume Screening Systems and Recruitment AI

6. SAGE Journals (2020). AI and discriminative decisions in recruitment: Challenging the core assumptions. [\[Link\]](#)

This paper reviews approaches to mitigating bias in automated recruitment systems, with a focus on NLP and ML models used for resume screening. It identifies common sources of bias, such as gendered wording in job descriptions or imbalanced training datasets. The paper categorizes mitigation techniques into pre-processing, in-processing, and post-processing strategies. Pre-processing includes data balancing, debiasing word embeddings, and anonymization of resumes. In-processing covers algorithmic changes. Post-processing involves adjusting decision thresholds or re-ranking outputs. The authors report that pre-processing and in-processing generally outperform post-processing. The paper concludes that hybrid approaches (combining pre-processing with in-processing) yield the most promising results for resume screening.

7. Journal of Business Ethics (2022). Ethics of AI-Enabled Recruiting and Selection: A Review and Research Agenda. [\[Link\]](#)

This paper is a systematic literature review of how artificial intelligence is used in recruitment and selection, with a focus on its ethical implications. Authors reviewed 51 relevant articles and covered AI use in all recruitment stages: outreach, screening, assessment, and facilitation. They also classified literature into five perspectives: Practitioner, Technical, Descriptive, Legal, and Theoretical. They found some research gaps present in the articles, such as lack of strong theoretical foundations, limited empirical evidence, and inconsistent fairness perception results.

8. Nature (2023). Ethics and Discrimination in AI Recruitment. [\[Link\]](#)

This paper is a literature review and empirical study on algorithmic discrimination in AI-enabled recruitment. It examines how AI tools can improve efficiency, recruitment quality, and decision-making, but also how they risk perpetuating bias in hiring based on gender, race, color, and even personality traits. The authors argue that these biases often originate from biased training datasets and human prejudice embedded in algorithm design. It then reviews 49 papers to categorize sources and types of discrimination, finding the literature lacking a unified framework. The paper concludes that mitigating bias requires both algorithmic improvements and stronger governance structures.

9. ScienceDirect (2024). Fairness, AI & Recruitment: A Scoping Review. [\[Link\]](#)

A scoping review of fairness in AI recruitment systems. The authors reviewed 106 peer-reviewed papers, categorizing them into three main themes: fairness definitions and metrics in recruitment, sources of bias, and bias mitigation strategies – both pre-processing, in-processing, and post-processing.

3 Demonstrated Bias in Existing Models

10. PACM HCI (CSCW) (2019). Discrimination through Optimization: How Facebook’s Ad Delivery Can Lead to Biased Outcomes. [Link]

This article investigates how Facebook’s ad delivery algorithms themselves, not just advertisers’ targeting choices, can cause discriminatory outcomes in online advertising. One of the key domains the authors tested was employment ads. They specifically ran real-looking job advertisements (e.g., for AI developers, nurses, supermarket cashiers, taxi drivers) and found that Facebook’s delivery optimization caused large gender and racial skews in who actually saw the ads, even though the targeting settings were identical and neutral. Employment and housing ads showed significant demographic skews — e.g., lumber jobs reached 90% men, cashier jobs 85% women, taxi jobs 75% Black users — despite identical targeting.

4 Proposed Solutions and Architectures

11. AIES (2018). Mitigating Unwanted Biases with Adversarial Learning. [Link]

This article presents a general framework for reducing demographic bias in machine learning models using adversarial training.

Method:

1. Train a predictor for the main task (e.g., predict income, complete an analogy).
2. Simultaneously train an adversary that tries to predict a protected variable (e.g., gender, race) from the predictor’s output.
3. Update the predictor to maximize task accuracy while minimizing the adversary’s success, thereby reducing bias.

This works for both regression and classification and is applicable to continuous or discrete outputs and protected variables.

12. Patterns (2024). FAIM: Fairness-aware Interpretable Modeling. [Link]

This paper presents *FAIM* (Fairness-Aware Interpretable Modeling), a framework designed to jointly optimize interpretability and fairness without significant loss of performance. The authors argue that traditional approaches treat fairness and interpretability as separate goals, often leading to trade-offs, whereas FAIM integrates them from the start.

The framework operates by:

1. Using inherently interpretable model classes (decision sets, rule lists).
2. Embedding fairness constraints directly into the model training process (demographic parity, equalized odds).
3. Leveraging a multi-objective optimization approach to balance accuracy, fairness, and simplicity.

Instead of the classical loss = error, they use:

$$\text{loss} = \text{error} + \lambda_1 \cdot \text{fairness penalty} + \lambda_2 \cdot \text{complexity penalty}$$

FAIM was tested on multiple datasets with protected attributes. It achieved fairness levels comparable to specialized debiasing algorithms. It retained high interpretability, allowing end-users to understand decisions. It also reduced the complexity of post-hoc explanations, due to the model’s inherently transparent decision logic.

13. arXiv (2019). Fairness in Deep Learning: A Computational Perspective. [Link]

The paper explains how to detect and mitigate discrimination in deep neural networks.

1. Problem: Even if explicit protected attributes (e.g., gender, age, race) are removed, models can *implicitly* encode them via correlated features (proxy variables), leading to *representation bias*. A second issue is *prediction quality disparity*, where performance metrics differ across demographic groups.
2. Interpretability: Use explanation methods (Integrated Gradients, SHAP, saliency/Grad-CAM, representation probing) to identify influential features and test whether protected information is encoded in intermediate representations.
3. Mitigation:
 - *Pre-processing*: data rebalancing/resampling, debiasing or removing proxy features, targeted augmentation.
 - *In-processing*: fairness-aware losses (e.g., demographic parity or equalized-odds penalties), adversarial debiasing, fair representation learning.
 - *Post-processing*: group-specific threshold adjustment, calibration, probabilistic relabeling/re-ranking to meet fairness constraints.
4. Conclusion: Most effective practice combines interpretability-driven *diagnosis* with targeted mitigation (often hybrid pre-/in-/post-processing), acknowledging trade-offs among fairness, accuracy, and other constraints.

14. NeurIPS (2016). Equality of Opportunity in Supervised Learning. [Link]

Authors introduce equalized odds and equal opportunity as fairness criteria for supervised ML, and a post-processing method to enforce them without retraining models.

Proposed fairness definitions:

1. Equalized Odds – Predictions should be independent of the protected attribute given the true outcome.
2. Equal Opportunity – A relaxation requiring only equal true positive rates across groups. The TPR should be the same across groups, but the FPR can differ.

The authors don’t change models. Instead, they take any model a person already has and apply a post-processing step to make it fair according to equalized odds or equal opportunity. The step is to find TPR and FPR for each group:

$$\text{TPR}(t) = \frac{\text{TP}(t)}{\text{TP}(t) + \text{FN}(t)}, \quad \text{FPR}(t) = \frac{\text{FP}(t)}{\text{FP}(t) + \text{TN}(t)}$$

Then, instead of using the same cutoff score for everyone, they set different thresholds per group so that the TPR/FPR match.

The output: for each group, a piecewise rule like (for instance):

1. If score is more than 0.82 $\rightarrow$ approve
2. If score is between 0.75 and 0.82 $\rightarrow$ approve with probability 0.5
3. Else $\rightarrow$ reject

15. UMAP (2020). Mitigating Demographic Bias in AI-based Resume Filtering [Link]

Authors collected 135 resumes from Chinese, Indian, and Malaysian candidates in Singapore, plus 9 finance/accounting job postings (3 from each nationality group). Using cosine similarity on tf-idf (Term Frequency-Inverse Document Frequency) vectors, the top-5 resumes per job were selected. Result:

1. Malaysia: 46.66% of matches
2. India: 42.22% of matches
3. China: 11.11% of matches

The standard tf-idf cosine similarity approach disproportionately favors certain nationalities. Visual t-SNE plots showed resumes clustered strongly by nationality, explaining low cross-national matches.

$$\text{TF-IDF}(t) = \text{TF}(t) \times \text{IDF}(t).$$

where

$$\begin{aligned} \text{TF}(t) &= \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}. \\ \text{IDF}(t) &= \log \left(\frac{\text{Total number of documents}}{\text{Number of documents containing term } t} \right). \end{aligned}$$

Proposed Method: fair-tf-idf

The authors introduce a reweighting factor $p\text{-ratio}$ to mitigate demographic bias in tf-idf representations:

$$p\text{-ratio}(t) = \frac{P(t \mid \text{lowest frequency demographic})}{P(t \mid \text{highest frequency demographic})}.$$

- If a term appears only in one demographic, $p\text{-ratio}(t) = 0$.
- If a term appears equally across demographics, $p\text{-ratio}(t) = 1$ and the weight remains unchanged.

The debiased weight is then:

$$\text{fair-tf-idf}(t) = \text{tf}(t) \times \text{idf}(t) \times p\text{-ratio}(t).$$

Improved Method: Fair-tf-idf with Sigmoid Transformation

To control the fairness-accuracy trade-off and avoid over-weighting common stopwords, we apply a sigmoid transformation:

$$\text{fair-tf-idf}(t) = \text{tf}(t) \times \text{idf}(t) \times \sigma(\lambda(p\text{-ratio}(t) - \tau)),$$

where the sigmoid function is

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

- $\lambda > 0$: controls the sharpness of the cut-off.
- $\tau \in (0, 1)$: sets the tolerance for demographic imbalance.

This preserves useful, widely shared terms while sharply down-weighting demographically skewed terms.

The results show that fair-tf-idf and fair-tf-idf + Sigmoid give much better results than the baseline tf-idf.

Method	Fairness $\uparrow$	Accuracy (%) $\uparrow$
tf-idf	0.0723	90.00
fair-tf-idf	0.3673	73.33
fair-tf-idf + Sigmoid	0.9230	86.66

5

16. Raghavan et al. (2019/2020). Mitigating Bias in Algorithmic Hiring: Evaluating Claims and Practices. [Link]

A landscape study of pre-employment assessment vendors: what they predict, how they validate, and how they claim to “debias.” It catalogs where bias creeps in (targets, features, validation) and maps techniques to U.S. anti-discrimination law—useful scaffolding for a resume-ranking system’s governance section.

17. Geyik, Ambler & Kenthapadi (2019). Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search. [Link]

This paper proposes complementary measures to quantify bias with respect to protected attributes such as gender and age and algorithms for computing fairness-aware re-ranking of results. Industrial framework to measure and re-rank candidate lists so exposure of protected groups aligns with policy. Includes offline/online results at LinkedIn.

18. Li, Raymond & Bergman (2025; NBER WP 2020). Hiring as Exploration. [Link]

Firms must balance “exploitation” with “exploration” when looking for candidates. Yet modern hiring algorithms are designed solely for exploitation. Authors of this article build a resume screening algorithm that values exploration by evaluating candidates according to their statistical upside potential.

19. Köchling & Wehner (2020). Discriminated by an Algorithm? Systematic Review in HR [Link]

Systematic review of discrimination and fairness in HR analytics, including screening tools. Summarizes where bias originates (data, labels, proxies) and which mitigations are credible in practice.

20. Zhang et al. (2022). Examining Perceptions Toward Hiring Algorithms. [Link]

A survey study on public attitudes toward algorithmic hiring. Overall, respondents judged hiring algorithms as neither fair nor effective. Perceptions varied by demographics: women

were less positive than men, and people with higher income and education reported more favorable views than others.

21. Peña et al. (2020). FairCVtest: Bias in Multimodal AI for Recruitment. [Link]

Authors trained automatic recruitment algorithms using a set of multimodal synthetic profiles consciously scored with gender and racial biases and presented a list of recent works developing techniques capable of removing sensitive information from the decision-making process of deep learning architectures..

22. Zhu et al. (2018). PJFNN: Person–Job Fit with Deep Neural Networks. [Link]

Early deep model that jointly encodes resumes and job descriptions to score fit.

23. Kleinberg, Mullainathan & Raghavan (2016). Inherent Trade-Offs in the Fair Determination of Risk Scores. [Link]

A seminal impossibility result proving that three natural desiderata — calibration within groups, balance for the positive class, and balance for the negative class — cannot be simultaneously satisfied except in special cases. The paper formalizes why many fairness debates deadlock: when base rates differ, you must pick which error disparate to tolerate.

24. Chouldechova (2017). Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. [Link]

Independently distills core tension between calibration and error-rate parity using criminal-justice data. Demonstrates bias towards criminal defendants mostly based on ethnicity.

25. Dwork, Hardt, Pitassi, Reingold & Zemel (2012). Fairness Through Awareness. [Link]

Introduces individual fairness—“treat similar individuals similarly”—via a task-specific similarity metric and gives optimization-based learning procedures that respect this Lipschitz-style constraint. Unlike group metrics, this protects individuals even within the same demographic, aligning with recruiter intuitions about comparable résumés. The paper also connects to privacy and metric design.

26. Kearns, Neel, Roth & Wu (2018). Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness. [Link]

Shows that satisfying fairness on a few coarse groups can still hide severe violations on subgroups. Proves convergence to the best subgroup-fair classifier. This motivates audits on rich, intersecting slices and supports using multicalibration or subgroup-aware constraints to avoid masking harms.

27. Kusner, Loftus, Russell & Silva (2017). Counterfactual Fairness. [Link]

Defines fairness using causal counterfactuals: "a decision is fair towards an individual if it the same in (a) the actual world and (b) a counterfactual world where the individual belonged to a different demographic group". Provides learning strategies and a case study (law school success).

28. Feldman, Friedler, Moeller, Scheidegger & Venkatasubramanian (2015). Certifying and Removing Disparate Impact. [Link]

Links the U.S. legal notion of disparate impact to measurable information leakage of protected attributes and proposes a data repair pre-processing transform that reduces leakage while preserving task signal. Authors propose a test for disparate impact based

on analyzing the information leakage of the protected class from the other data attributes and provide methods to make data unbiased. Empirically shows improved fairness with strong utility retention by “fixing the data”.

29. Rudin (2019). Stop Explaining Black Box Models for High-Stakes Decisions and Use Interpretable Models Instead. [Link]

Argues that post-hoc explanations can mislead and that for consequential domains (hiring), inherently interpretable models should be preferred.

30. Doshi-Velez & Kim (2017). Towards a Rigorous Science of Interpretable Machine Learning. [Link]

A position paper laying out a taxonomy of interpretability evaluations: application-grounded (real users & tasks), human-grounded (proxy tasks & real humans), and functionally-grounded. It clarifies when interpretability is required.

31. Hashimoto, Srivastava, Namkoong & Liang (2018). Fairness Without Demographics in Repeated Loss Minimization. [Link]

Minority groups contribute less to the training objective and thus tend to suffer higher loss. Worse, as model accuracy affects user retention, a minority group can shrink over time. Authors first show that the status quo of empirical risk minimization (ERM) amplifies representation disparity over time, which can even make initially fair models unfair. To mitigate this, they develop an approach based on distributionally robust optimization (DRO). Researchers prove that this approach controls the risk of the minority group at each time step.

32. González Sendino, R., Serrano, E., Bajo, J., and Novais, P. (2024). A Review of Bias and Fairness in Artificial Intelligence. *International Journal of Interactive Multimedia and Artificial Intelligence*, 9(1), 5–17. [Link]

This paper contributes with: (1) a categorization of biases; (2) a revision of fairness metrics to audit the data and AI models; and, (3) a novel taxonomy of the procedures to mitigate biases in the different phases of an AI model’s development.

33. Köchling, A., Wehner, M.C. Discriminated by an algorithm: a systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development. *Bus Res* 13, 795–848 (2020). [Link]

A systematic review of 36 journal articles from 2014 to 2020. Authors present some applications of algorithmic decision-making and evaluate the possible pitfalls in these two essential HR functions.

34. Fernández-Martínez, Carmen and Fernández, Alberto. "AI and recruiting software: Ethical and legal implications" *Paladyn, Journal of Behavioral Robotics*, vol. 11, no. 1, 2020, pp. 199-216. [Link]

This article examines the state-of-the-art applications of AI in human resources, focusing on the ethical and legal consequences of using AI in recruitment processes. It discusses issues such as discrimination based on gender and race and emphasizes the need for external and neutral auditing of AI systems.

35. ...

36. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. Marco Tulio Ribeiro, Sameer Singh, Carlos Guestrin. [Link]

LIME

37. A Unified Approach to Interpreting Model Predictions. Scott Lundberg, Su-In Lee. [\[Link\]](#)

SHAP

38. Axiomatic Attribution for Deep Networks. Mukund Sundararajan, Ankur Taly, Qiqi Yan. [\[Link\]](#)

Integrated Gradients

39. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, Dhruv Batra. [\[Link\]](#)

GradCAM

40. Learning Important Features Through Propagating Activation Differences. Avanti Shrikumar, Peyton Greenside, Anshul Kundaje. [\[Link\]](#)

DeepLIFT

6 Datasets

Surendra Kumar Nellore. Recruitment Dataset [\[Link\]](#)

Suriya Ganesh. 54k Resume dataset (structured) [\[Link\]](#)

resumes dataset by datasetmaster [\[Link\]](#)

Resume dataset by brackozi [\[Link\]](#)

Snehaan Bhawal. Resume Dataset [\[Link\]](#)

Dmitriy Vasilev. HH resumes analysis [\[Link\]](#)

58k vacancies dataset [\[Link\]](#)